

5.4 Green's Theorem In-Class Practice

1. Consider the region R bounded by the curves $y = x^2$ and $y = \sqrt[3]{x}$ in the first quadrant. Suppose that the boundary of this region is the curve C oriented as shown in the image below.
- a) Use Green's Theorem to determine $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle xy, y^2 + x \rangle$. (Be sure to show that all conditions are met.)

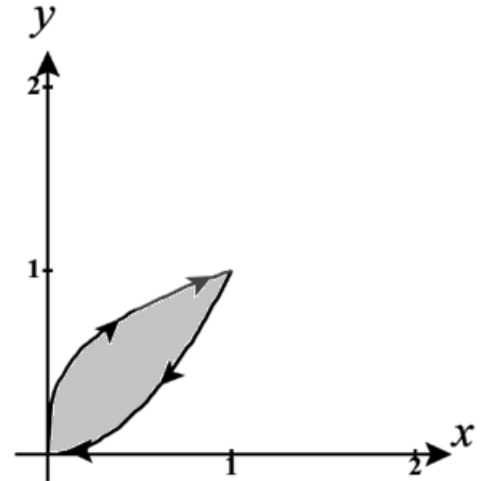
We see that C is a piecewise smooth, simple, closed curve and $\mathbf{F}$ has continuous partials everywhere.

We note that C is not positively oriented, but instead we use $-C$.

We also calculate that $\frac{\partial P}{\partial y} = x$ & $\frac{\partial Q}{\partial x} = 1$.

We also define R as $R = \{(x, y) \mid x^2 \leq y \leq \sqrt[3]{x} \text{ \& } 0 \leq x \leq 1\}$

Therefore, by Green's Theorem,



$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_0^1 \int_{x^2}^{\sqrt[3]{x}} (x-1) dy dx = \int_0^1 (x-1) y \Big|_{x^2}^{\sqrt[3]{x}} dx = \int_0^1 (x-1) (\sqrt[3]{x} - x^2) dx \\ &= \int_0^1 (x^{4/3} - x^3 - \sqrt[3]{x} + x^2) dx = \left(\frac{3}{7} x^{7/3} - \frac{1}{4} x^4 - \frac{3}{4} x^{4/3} + \frac{1}{3} x^3 \right) \Big|_0^1 = \left(\frac{3}{7} - \frac{1}{4} - \frac{3}{4} + \frac{1}{3} \right) - (0) \\ &= \frac{36}{84} - \frac{84}{84} + \frac{28}{84} = -\frac{20}{84} = -\frac{5}{21} \end{aligned}$$

- b) Calculate $\int_C \mathbf{F} \cdot d\mathbf{r}$ directly (that is, without the use of Green's Theorem)

Let C_1 be the curve for $y = x^2$ where $0 \leq x \leq 1$. We parameterize this curve as $\mathbf{r}_1(t) = \langle 1-t, (1-t)^2 \rangle$ where $0 \leq t \leq 1$.

Let C_2 be the curve for $y = \sqrt[3]{x}$ where $0 \leq x \leq 1$. We parameterize this curve as $\mathbf{r}_1(t) = \langle t, \sqrt[3]{t} \rangle$ where $0 \leq t \leq 1$.

We then see that

$$\begin{aligned}
\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} \\
&= \int_0^1 \langle xy, y^2 + x \rangle \cdot \langle dx, dy \rangle + \int_0^1 \langle t^{4/3}, t^{2/3} + t \rangle \cdot \left\langle 1, \frac{1}{3}t^{-1/3} \right\rangle dt \\
&= \int_0^1 \langle xy, y^2 + x \rangle \cdot \langle dx, dy \rangle + \int_0^1 \langle t^{4/3}, t^{2/3} + t \rangle \cdot \left\langle 1, \frac{1}{3}t^{-1/3} \right\rangle dt \\
&= \int_0^1 \langle (1-t)^3, (1-t)^4 + (1-t) \rangle \cdot \langle -1dt, -2(1-t)dt \rangle + \int_0^1 \langle t^{4/3}, t^{2/3} + t \rangle \cdot \left\langle 1, \frac{1}{3}t^{-1/3} \right\rangle dt \\
&= \int_0^1 \left(-(1-t)^3 + 2t(t^4 + t) \right) dt + \int_0^1 \left(t^{4/3} + \frac{1}{3}t^{-1/3}(t^{2/3} + t) \right) dt \\
&= \int_0^1 (t^3 + 2t^5 + 2t^2) dt + \int_0^1 \left(t^{4/3} + \frac{1}{3}t^{1/3} + \frac{1}{3}t^{2/3} \right) dt \\
&= \left(\frac{1}{4}t^4 + \frac{1}{3}t^6 + \frac{2}{3}t^3 \right) \Big|_0^1 + \left(\frac{3}{7}t^{7/3} + \frac{1}{4}t^{4/3} + \frac{1}{5}t^{5/3} \right) \Big|_0^1 \\
&= \left(\frac{1}{4} + \frac{1}{3} + \frac{2}{3} \right) + \left(\frac{3}{7} + \frac{1}{4} + \frac{1}{5} \right) \\
&= \left(\frac{5}{4} \right) + \left(\frac{60}{140} + \frac{35}{140} + \frac{28}{140} \right) \\
&= \frac{175}{140} + \frac{123}{140} \\
&= \frac{298}{140}
\end{aligned}$$

2. Something moving counterclockwise
3. Something were they convert double integral into two line integrals.
4. Something atypical, like exam question for this year.

1. Consider the work done by the vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ on a particle moving counterclockwise around the unit circle.

a) Use Green's Theorem to evaluate the resulting line integral.

A picture of the curve C and implied region D are shown at the right.

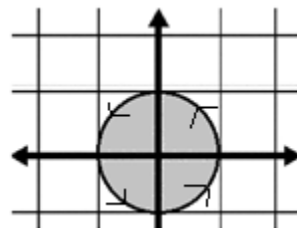
$$\text{Here, } D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq 1\}.$$

We note that $P(x, y) = y$ \& $Q(x, y) = -x$ and so $\frac{\partial P}{\partial y} = 1$ \& $\frac{\partial Q}{\partial x} = -1$.

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .

Therefore, by Green's Theorem,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C ydx - xdy = \iint_D (-1 - 1) dA = \iint_D -2dA = -2 \iint_D dA = -2\pi$$



- b) How would your answer change if the particle was moving clockwise around the unit circle?

Moving in the clockwise direction would change our path C to $-C$. Thus, the only difference would be that the total work would be multiplied by negative one.

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = - \int_C \mathbf{F} \cdot d\mathbf{r} = - \int_C ydx - xdy = - \iint_D (-1 - 1) dA = \iint_D -2dA = 2 \iint_D dA = 2\pi.$$

2. Use Green's Theorem to evaluate the line integral where $\mathbf{F} = xy\mathbf{i} + (x - y)\mathbf{j}$ and C is the triangle joining $(2, 0)$, $(0, 2)$, and $(-2, 0)$ in the counterclockwise direction.

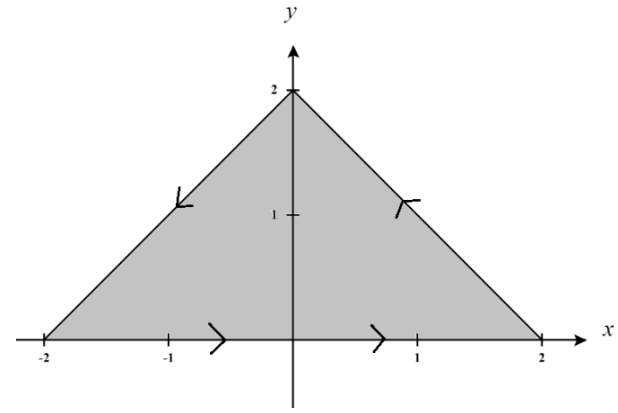
A picture of the curve C and implied region D are shown at the right.

Here, $D = \{(x, y) | y - 2 \leq x \leq 2 - y \text{ \& } 0 \leq y \leq 2\}$.

We note that $P(x, y) = xy$ \& $Q(x, y) = x - y$ and so

$$\frac{\partial P}{\partial y} = x \text{ \& } \frac{\partial Q}{\partial x} = 1.$$

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .



Therefore, by Green's Theorem,

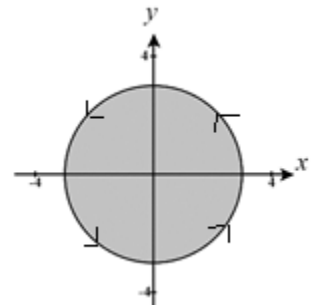
$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C xy dx + (x - y) dy = \iint_D (1 - x) dA = \int_0^2 \int_{y-2}^{2-y} (1 - x) dA = \int_0^2 \left(x - \frac{1}{2} x^2 \right) \Big|_{y-2}^{2-y} dy \\ &= \int_0^2 \left[\left((2 - y) - \frac{1}{2} (2 - y)^2 \right) - \left((y - 2) - \frac{1}{2} (y - 2)^2 \right) \right] dy \\ &= \int_0^2 \left[2 - y - 2 + 2y - \frac{1}{2} y^2 - \left(y - 2 - \frac{1}{2} y^2 + 2y - 2 \right) \right] dy \\ &= \int_0^2 \left[2 - y - 2 + 2y - \frac{1}{2} y^2 - y + 2 + \frac{1}{2} y^2 - 2y + 2 \right] dy \\ &= \int_0^2 [4 - 2y] dy \\ &= (4y - y^2) \Big|_0^2 \\ &= (8 - 4) + (0) \\ &= 4 \end{aligned}$$

3. Evaluate $\oint_C (x^2 + y^2) dx + x dy$, where C is the circle $x^2 + y^2 = 9$.

A picture of the curve C and implied region D are shown at the right.

Here, $D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq 3\}$.

We note that $P(x, y) = x^2 + y^2$ \& $Q(x, y) = x$ and so $\frac{\partial P}{\partial y} = 2y$ \& $\frac{\partial Q}{\partial x} = 1$.



We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .

Therefore, by Green's Theorem,

$$\begin{aligned}\oint_C (x^2 + y^2) dx + x dy &= \iint_D (1 - 2y) dA = \int_0^{2\pi} \int_0^3 (1 - 2r \sin(\theta)) r dr d\theta = \int_0^{2\pi} \int_0^3 (r - 2r^2 \sin(\theta)) dr d\theta \\ &= \int_0^{2\pi} \left[\frac{1}{2} r^2 - \frac{2}{3} r^3 \sin(\theta) \right]_0^3 d\theta = \int_0^{2\pi} \left[\frac{9}{2} - 18 \sin(\theta) \right] d\theta \\ &= \left[\frac{9}{2} \theta + 18 \cos(\theta) \right]_0^{2\pi} = [9\pi + 18] - [0 + 18] \\ &= 9\pi\end{aligned}$$

4. Suppose an object is being spun by high wind currents modelled by the vector field

$$\mathbf{F}(x, y) = \left\langle -y^3 + x \tan^{-1}(y), x^3 + 2 + \frac{x^2}{2 + 2y^2} \right\rangle \text{ in a counterclockwise fashion around a circle of radius 2.}$$

Determine the total work done by the wind currents on the object as the object moves around the circle once.

A picture of the curve C and implied region D are shown at the right.

$$\text{Here, } D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq 2\}.$$

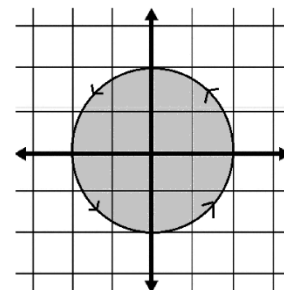
We note that $P(x, y) = -y^3 + x \tan^{-1}(y)$ \& $Q(x, y) = x^3 + 2 + \frac{x^2}{2 + 2y^2}$ and

$$\text{so } \frac{\partial P}{\partial y} = -3y^2 + \frac{x}{1 + y^2} \text{ \& } \frac{\partial Q}{\partial x} = 3x^2 + \frac{x}{1 + y^2}.$$

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .

Therefore, by Green's Theorem,

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C (-y^3 + x \tan^{-1}(y)) dx + \left(x^3 + 2 + \frac{x^2}{2 + 2y^2} \right) dy = \iint_D \left[\left(3x^2 + \frac{x}{1 + y^2} \right) - \left(-3y^2 + \frac{x}{1 + y^2} \right) \right] dA \\ &= \iint_D (3x^2 + 3y^2) dA = \int_0^{2\pi} \int_0^2 3r^2 \cdot r dr d\theta \\ &= \int_0^{2\pi} \left[\frac{3}{4} r^4 \right]_0^2 d\theta = \int_0^{2\pi} 12 d\theta = 24\pi\end{aligned}$$



5. Determine the work done by the vector field $\mathbf{F}(x, y) = \langle e^{\cosh(x)} + 3y, e^{\sinh(y)} - 3x \rangle$ on a particle moving along C , where C is the boundary of the region outside the unit square but inside the circle $4x^2 + 4y^2 = 16$ with positive orientation.

A picture of the curve C and implied region D are shown at the right.

We note that $P(x, y) = e^{\cosh(x)} + 3y$ & $Q(x, y) = e^{\sinh(y)} - 3x$
and so $\frac{\partial P}{\partial y} = 3$ & $\frac{\partial Q}{\partial x} = -3$.

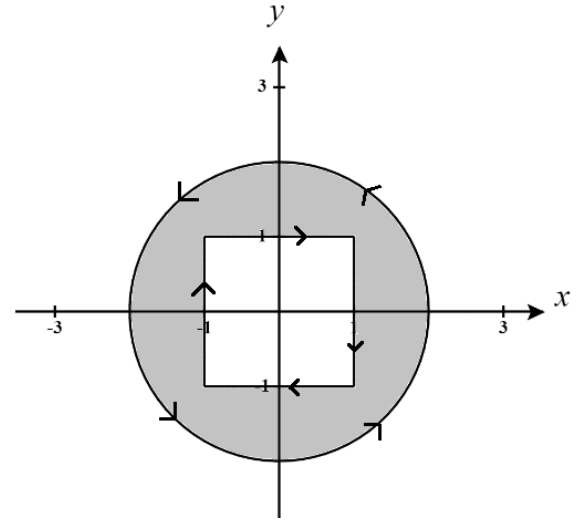
We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .

Therefore, by Green's Theorem,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (e^{\cosh(x)} + 3y)dx + (e^{\sinh(y)} - 3x)dy = \iint_D (-3 - 3)dA = \iint_D -6dA = -6 \iint_D dA$$

We know from basic geometry that the area of the region is given by $A = \pi(2)^2 - 1 = 4\pi - 1$

$$\text{Thus, } \int_C \mathbf{F} \cdot d\mathbf{r} = -6 \iint_D dA = -6(4\pi - 1) = -24\pi + 6$$



6. Use Green's Theorem to evaluate the line integral along the given positively oriented curve.

$$\int_C (2x + \cos(y^2))dy + (y + e^{\sqrt{x}})dx$$

where C is the boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.

A picture of the curve C and implied region D are shown at the right.

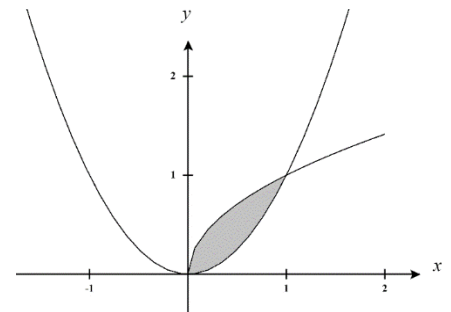
$$\text{Here, } D = \{(x, y) | x^2 \leq y \leq \sqrt{x} \text{ \& } 0 \leq x \leq 1\}.$$

We note that $P(x, y) = y + e^{\sqrt{x}}$ & $Q(x, y) = 2x + \cos(y^2)$ and so

$$\frac{\partial P}{\partial y} = 1 \text{ \& } \frac{\partial Q}{\partial x} = 2.$$

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .

Therefore, by Green's Theorem,



$$\begin{aligned}
\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C (2x + \cos(y^2)) dy + (y + e^{\sqrt{x}}) dx = \iint_D (2-1) dA = \iint_D dA \\
&= \int_0^1 \int_{x^2}^{\sqrt{x}} dy dx = \int_0^1 (y) \Big|_{x^2}^{\sqrt{x}} dx \\
&= \int_0^1 (\sqrt{x} - x^2) dx = \left(\frac{2}{3} x^{3/2} - \frac{1}{3} x^3 \right) \Big|_0^1 \\
&= \left(\frac{2}{3} - \frac{1}{3} \right) = \frac{1}{3}
\end{aligned}$$

7. Explain why green's theorem cannot be used to evaluate the following line integral:

$$\int_C \ln(1+2y) dx + \frac{3}{2} \left(x - \frac{1}{2} \right)^{2/3} dy \text{ where } C \text{ is the boundary of positively oriented unit circle.}$$

A picture of the curve C and implied region D are shown at the right.

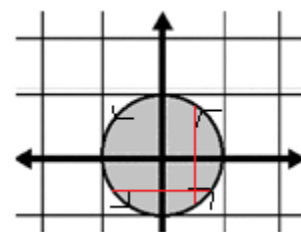
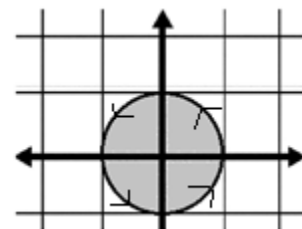
$$\text{Here, } D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq 1\}.$$

We note that $P(x, y) = \ln(1+2y)$ \& $Q(x, y) = \frac{3}{2} \left(x - \frac{1}{2} \right)^{2/3}$ and so

$$\frac{\partial P}{\partial y} = \frac{2}{1+2y} \text{ \& } \frac{\partial Q}{\partial x} = \frac{1}{\sqrt[3]{x - \frac{1}{2}}}.$$

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane. However, we see that $\frac{\partial P}{\partial y}$ is discontinuous when $y = -\frac{1}{2}$ and $\frac{\partial Q}{\partial x}$ is discontinuous

when $x = \frac{1}{2}$. Therefore, the first partials are not continuous on D . In fact, the first partials at all points shown in red in the picture to the right.



8. Compute $\int_C xe^{-2x} dx + (x^4 + 2x^2y^2 + y^4) dy$, where C is the negatively oriented boundary of the region enclosed by $y = x^2$, the x -axis, and $x = 2$.

A picture of the curve C and implied region D are shown at the right.

$$\text{Here, } D = \{(x, y) | 0 \leq y \leq x^2 \text{ \& } 0 \leq x \leq 2\}.$$

We note that

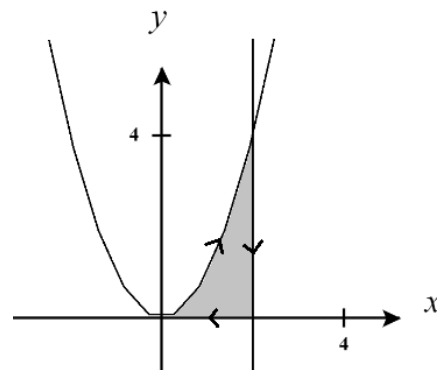
$$P(x, y) = xe^{-2x} \text{ \& } Q(x, y) = x^4 + 2x^2y^2 + y^4 = (x^2 + y^2)^2 \text{ and so}$$

$$\frac{\partial P}{\partial y} = 0 \text{ \& } \frac{\partial Q}{\partial x} = 4x^3 + 4xy^2 = 4x(x^2 + y^2).$$

We see that $-C$ is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .

Therefore, by Green's Theorem,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= - \int_{-C} \mathbf{F} \cdot d\mathbf{r} = - \int_{-C} (xe^{-2x}) dx + (x^4 + 2x^2y^2 + y^4) dy = - \iint_D (4x^3 + 4xy^2) dA \\ &= - \int_0^2 \int_0^{x^2} (4x^3 + 4xy^2) dy dx = - \int_0^2 \left(4x^3y + \frac{4}{3}xy^3 \right) \Big|_0^{x^2} dx \\ &= - \int_0^2 \left(4x^5 + \frac{4}{3}x^7 \right) dx = - \left(\frac{2}{3}x^6 + \frac{1}{6}x^8 \right) \Big|_0^2 \\ &= - \left(\frac{2}{3}(64) + \frac{1}{6}(256) \right) \\ &= - \frac{256}{3} \end{aligned}$$



9. Use Green's Theorem to evaluate the following line integral where C is the path from $(1,0)$ to $(3,0)$ along a line, followed by the path from $(3,0)$ to $(-3,0)$ along the top half of a circle of radius 3 centered at the origin, followed by the path from $(-3,0)$ to $(-1,0)$ along a line, followed by the path from $(-1,0)$ to $(1,0)$ along the top half of a circle of radius 1 centered at the origin.

$$\int_C (\tan^{-1}(x) + y^2) dx + (e^y - x^2) dy$$

A picture of the curve C and implied region D are shown at the right.

$$\text{Here, } D = \{(r, \theta) | 0 \leq \theta \leq \pi \text{ \& } 1 \leq r \leq 3\}.$$

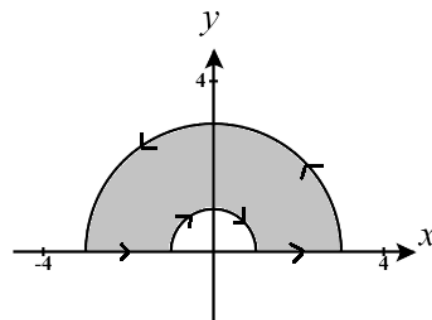
We note that $P(x, y) = \tan^{-1}(x) + y^2$ \& $Q(x, y) = e^y - x^2$ and so

$$\frac{\partial P}{\partial y} = 2y \text{ \& } \frac{\partial Q}{\partial x} = -2x.$$

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .

Therefore, by Green's Theorem,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C (\tan^{-1}(x) + y^2) dx + (e^y - x^2) dy = \iint_D (-2x - 2y) dA \\ &= -2 \iint_D (x + y) dA = -2 \int_0^\pi \int_1^3 (r \cos(\theta) + r \sin(\theta)) \cdot r dr d\theta \\ &= -2 \int_0^\pi \int_1^3 r^2 [\cos(\theta) + \sin(\theta)] dr d\theta = -2 \int_0^\pi \frac{1}{3} r^3 [\cos(\theta) + \sin(\theta)] \Big|_1^3 d\theta \\ &= -2 \int_0^\pi \frac{26}{3} [\cos(\theta) + \sin(\theta)] d\theta = -\frac{52}{3} \int_0^\pi [\cos(\theta) + \sin(\theta)] d\theta \\ &= -\frac{52}{3} [\sin(\theta) - \cos(\theta)] \Big|_0^\pi = -\frac{52}{3} [(0+1) - (0-1)] = -\frac{52}{3} [2] \\ &= -\frac{104}{3} \end{aligned}$$



10. Evaluate $\oint_C (x^{\sin(x)} - x - 2y) dx + (2x - \tan^{-1}(y) - y^{\cos(y)}) dy$, where C is the ellipse $x^2 + 4y^2 = 16$.

A picture of the curve C and implied region D are shown at the right.

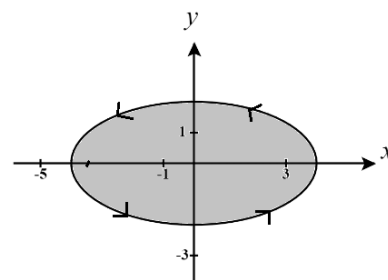
We note that

$P(x, y) = x^{\sin(x)} - x - 2y$ \& $Q(x, y) = 2x - \tan^{-1}(y) - y^{\cos(y)}$ and so

$$\frac{\partial P}{\partial y} = -2 \text{ \& } \frac{\partial Q}{\partial x} = 2.$$

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy -plane and that P and Q have continuous first order partial derivatives on D .

Therefore, by Green's Theorem,



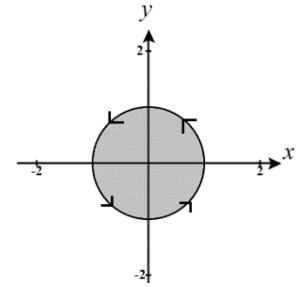
$$\oint_C \left(x^{\sin(x)} - x - 2y \right) dx + \left(2x - \tan^{-1}(y) - y^{\cos(y)} \right) dy = \iint_D (2 - (-2)) dA = \iint_D 4 dA = 4 \iint_D dA$$

To calculate this more easily we will use a change of variables to rework the region D into the unit circle.

Let $x = 4u$ & $y = 2v$ then we see that S is determined by

$$\begin{aligned} x^2 + 4y^2 = 16 &\Rightarrow (4u)^2 + 4(2v)^2 = 16 \\ &\Rightarrow 16u^2 + 16v^2 = 16 \Rightarrow u^2 + v^2 = 1 \end{aligned}$$

So, $S = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& \; } 0 \leq r \leq 1\}$ and has the following picture



Therefore, our transformation is $T(u, v) = (4u, 2v)$ and the Jacobian of this transformation is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} = 8 - 0 = 8$$

Thus,

$$\begin{aligned} \oint_C \left(x^{\sin(x)} - x - 2y \right) dx + \left(2x - \tan^{-1}(y) - y^{\cos(y)} \right) dy &= 4 \iint_D dA \\ &= 4 \iint_S |8| du dv \\ &= 32 \int_0^{2\pi} \int_0^1 r dr d\theta \\ &= 32 \int_0^{2\pi} \frac{1}{2} d\theta \\ &= 32\pi \end{aligned}$$

NOTE: You may notice that in general the area of an ellipse is given by $A = ab\pi$ where a & b are the radii in each direction of the ellipse. In the example above we have $A = (4)(2)\pi$ and thus

$$4 \iint_D dA = 4 \cdot 8\pi = 32\pi$$